Tejas Singh

+1 778-881-4645 | tejasxsingh@gmail.com | github.com/tejasxsingh | linkedin.com/in/tejassingh23 | tejasxsingh.github.io

Education

University of British Columbia — Vancouver, BC
BSc Mathematics | ***$80,000** International Major Entrance Scholarship* — *Sep 2022 – Present*
– **Coursework:** Machine Learning, Statistical Inference, Linear Models, Data Structures & Algorithms, Software Engineering

Machine Learning & Software Engineering Experience

AI Developer — Sep 2026 – Present
BC Cancer — *Vancouver, BC*
– **RAG & NLP Evaluation:** Develop a **Python** evaluator using **100 standardized Q&A pairs** and expert answers, standardizing offline evaluation and error analysis for clinical reviewers.
– **Experimentation:** Run **4 evaluator types** to produce up to **400 scores per model version**, helping reviewers isolate retrieval relevance, grounding, citation accuracy, and completeness errors across **5 safety dimensions**.
– **Review Workflows:** Flag safety failures and route uncertain outputs for expert review across **4 delivery channels**, helping the team identify regressions before release.

Computer Vision & Machine Learning Engineer Intern — May 2026 – Aug 2026
Genist Systems — *Vancouver, BC*
– **Deep Learning:** Fine-tuned detection and transformer re-identification in **PyTorch**; tuned learning rates, augmentation, and thresholds using Bayesian optimization, evaluating precision, recall, and identity retention.
– **Inference Optimization:** Reduced end-to-end latency by **32%** through Python pipeline profiling and ONNX export, shortening the time from video input to detection output.
– **Production Reliability:** Cut deployment failures by **28%** through containerized inference and motion, lighting, and occlusion tests, making model delivery more reliable for the engineering team.

Quantitative Analyst - Machine Learning — Jan 2026 – Apr 2026
UBC Sauder | Connor, Clark & Lunn — *Vancouver, BC*
– **Financial Data:** Turned **54M+ observations** across **17,000 equities** into **1.47M reproducible training inputs** using vectorized **Python, NumPy, and pandas**, giving researchers repeatable inputs for model experiments.
– **Predictive Modeling:** Trained a ViT-Base/32 for **20-day prediction**; engineered stock-split-corrected labels and compared signals with statistical baselines to assess predictive value across market regimes.
– **Statistical Validation:** Tested signals across **30 years of unseen periods** and communicated results to **15+ reviewers**, translating model performance into evidence for research decisions.

Machine Learning Engineer — Sep 2025 – Apr 2026
Providence Health Care | UBC Computational Biology Lab — *Vancouver, BC*
– **Feature Engineering:** Aligned **3 biological modalities** spanning **80K+ cells** and **50K+ features**; applied **PCA and feature selection** to reduce noise in high-dimensional model inputs.
– **Research Implementation:** Built reusable **PyTorch** modules with configurable dimensions and sparsity; contributed to the smbpls package so researchers could apply the method to their own multi-modal datasets.
– **Model Validation:** Ran **4 ablations** and **50 experiments**, achieving held-out R^2 of **0.993/0.997**; compared feature and modality choices to guide researchers toward stronger model configurations.

Software Engineer — Sep 2025 – Dec 2025
CoDHerS, UBC Asian Centre — *Vancouver, BC*
– **Backend Engineering:** Reduced manual content processing by **40%** with PostgreSQL schemas and **10+ Python REST APIs**, automating validation, metadata handling, and search for research workflows.
– **Engineering Quality:** Collaborated with a **10-person team** through Git reviews; added request validation, tests, error handling, and documentation to support maintainable backend development.

Software Engineer, Technology Analytics — Sep 2024 – Dec 2024
Nishan Transport Inc. — *Dorval, QC*
– **Data Analysis:** Cut reporting time by **22%** by joining fleet, shipment, revenue, and cost data in **Python/SQL** and automating checks, giving operations staff faster access to consistent reports.
– **Decision Support:** Analyzed utilization, route costs, and delay distributions; automated KPI dashboards, improving reporting visibility by **30%** and helping stakeholders identify operational bottlenecks.

Selected Projects

Adaptive Factor Model with Regime Detection
Python, R, hmmlearn, statsmodels, SciPy, pandas, Seaborn
– **Regime Modeling:** Built Gaussian HMM and regime-conditional factor models on **simulated returns**, making changes in market, value, and momentum exposures visible across volatility regimes.
– **Statistical Diagnostics:** Paired Newey–West errors, Chow tests, and residual diagnostics with Seaborn visualizations and confidence intervals to help distinguish regime differences from estimation uncertainty.

RAG Research Assistant
Python, FastAPI, ChromaDB, LangChain, OpenAI API, Docker, Pytest
– **AI Application:** Automated **5 stages** from PDF parsing to citation-grounded answers, letting users query papers through FastAPI; mocked model calls so API tests run without paid inference.

Technical Skills

Programming & Systems: Python, SQL, Bash scripting, Unix/Linux, PostgreSQL, C++, R, data structures and algorithms
Applied ML: PyTorch, scikit-learn, deep learning, reinforcement learning, Transformers, feature engineering, PCA
Data & Visualization: NumPy, pandas, Seaborn, Streamlit, statsmodels, SciPy, OpenCV, KPI dashboards
Experimentation: MLflow, loss-function design, Bayesian optimization, cross-validation, ablations, error analysis
Statistics: Statistical inference, probabilistic modeling (HMM), regression, hypothesis testing, confidence intervals
Engineering: Networking, HTTP/REST APIs, FastAPI, Git, Docker, Pytest, ONNX, AWS, NVIDIA GPUs, RAG